

ETHAN MIKOLAYCIK

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EDUCATION

Texas A&M University

Bachelor of Science, Computer Science; Intended Minor in Electrical Engineering

College Station, TX

Aug 2025 – Expected 2029

EXPERIENCE

Software Lead

Sep 2025 – Present

WHOOOP VEX U Robotics Team

College Station, TX

- Manage and mentor software members, delegating them projects: LVGL PID auto-tuner, LIDAR MCL localization simulator, velocity controller for holonomic drivetrains, logger, limelight localization
- Rewrote our team's ReveilLib library to include path following, holonomic motions, ray-casting localization, gain scheduling, and a universal chassis controller
- Currently working on serial communication between Raspberry Pi 5 and VEX Brain with ROS2 for external sensors
- Built WHOOPSim which allows drivers to control a simulated robot with a VEX controller

PROJECTS

mikLib | C++, VEXcode

- Open-source library for VEXcode adding odometry, motion control, and an on-brain custom UI framework for real-time PID tuning and autonomous selection
- Implemented two-tracker IMU odometry, wall resetting, with turn, swing and drive to point, drive to pose, and holonomic motions
- Implemented reusable PID class and motion chaining, which values can be individually passed into each drive motion; SD-card save/load for tuning constants and selected autonomous
- Documented with setup tutorials, Doxygen, and a Docusaurus website; actively maintained and updated
- 600+ downloads on Github, with discord community for support and development

Wall Reset Simulator | Javascript, HTML, CSS

- Built a browser-based simulator to validate a wall-reset localization algorithm, using virtual lasers to estimate robot pose from distances to field walls
- Implemented configurable front/left/right/rear lasers with sensor offsets, live visibility toggles, and field switching
- Deployed on GitHub Pages with documentation

Scene Maker | Java AWT/SWING

- Built a customizable scene maker; scene is built with background and particles, with velocity, size and amount parameters; snowflake and rain particles are pre-built
- Snowflakes are procedurally generated, and can be selected to rotate along the x, y and z axes
- Placed 1st in CS3 winter scene competition

mikGen | React, TypeScript, Tailwind

- Open-source web-based path planner, autonomous simulator, and code generator for VEX Robotics, supports mikLib, ReveilLib, LemLib, JAR-Template, EZ-Template and custom libraries
- Segment-based path building (move to point/pose, turn/swing, drive/strafe) with animated robot playback that runs each library's real constants and backend for accurate simulation
- Library-specific code export/import via clipboard or file, undo/redo with auto-save, box selection, snap-to-grid and magnet snapping, ghost robots, and numbered paths
- Built as a React and TypeScript offline PWA, with a Docusaurus getting-started guide

SKILLS

Programming Languages C++, Javascript, Typescript, Java, Python, C#

Technologies Git, GitHub, Linux, CMake, React, Vite, Tailwind, Unity, ROS2, VS Code, Fusion 360

Interests Mountain biking, Lego, Origami